

POWER ON BESS 59MW/120MWH

PROJECT PROPOSAL

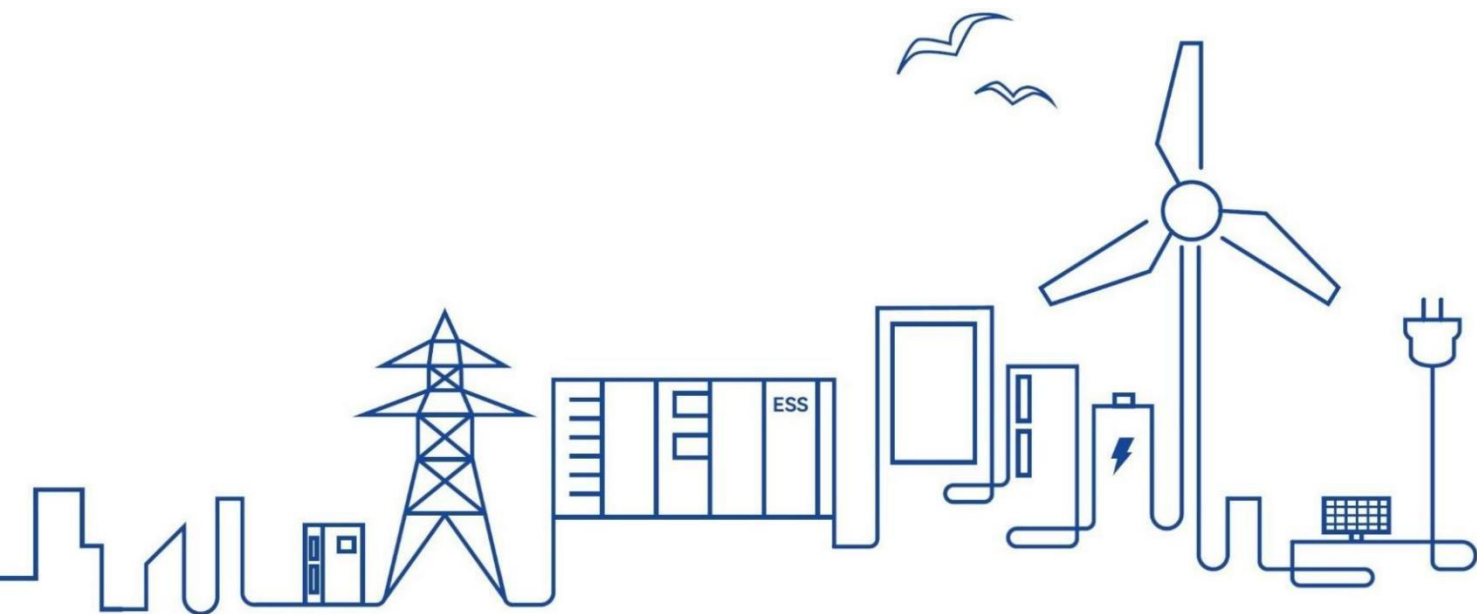


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● Abbreviations

BESS	Battery Energy Storage System
BMS	Battery Management System
BMU	Battery Management Unit
DC	Direct Current
DOD	Depth of Discharge
EMS	Energy Management System
FACP	Fire Alarm Control Panel
FSS	Fire Suppression System
HVAC	Heating Ventilation and Air Conditioning
POC	Point of Connection
SOC	State of Charge
SOH	State of Health
RSOC	Resting SOC
RTE	Round-trip Efficiency

1 Project Summary and System Design

1.1 Project Summary

Table Project Information

No.	Project Information	
1	Project Address	Cyprus
2	Project Scale	60MW/120MWh initially, and a 12.5MW/25MWh for augmentation.
3	Point of Connection Voltage and Frequency	33kV@MV, 132kV@HV
4	Application Scenarios	Grid-following

1.2 System Configuration

The Project nameplate capacity is 60MW/120MWh initially, and a 12.5MW/25MWh for augmentation. The whole project consists of 29 sets of 5.015MWh liquid cooling battery container, 7 sets of 10MW PCS & MV skid container, 1 set of 2.5MW PCS & MV skid container.

Each 5.015MWh liquid cooling battery container is a 20 ft. HC container. which includes a 416S12P high security LFP battery system, liquid cooling system, a fire suppression system, and other auxiliary equipment.

Each 10MW PCS & MV skid container is a 40 ft. HC container, which contains eight PCS units with the power of 1250kW, one 10000kVA oil-immersed step-up transformer, cooling system and other auxiliary equipment. Each 2.5MW PCS & MV skid container is a 20 ft. HC container, which contains two PCS units with the power of 1250kW, one 2500kVA oil-immersed step-up transformer, cooling system and other auxiliary equipment.

1.3 System Diagram

System diagram is shown in Fig.

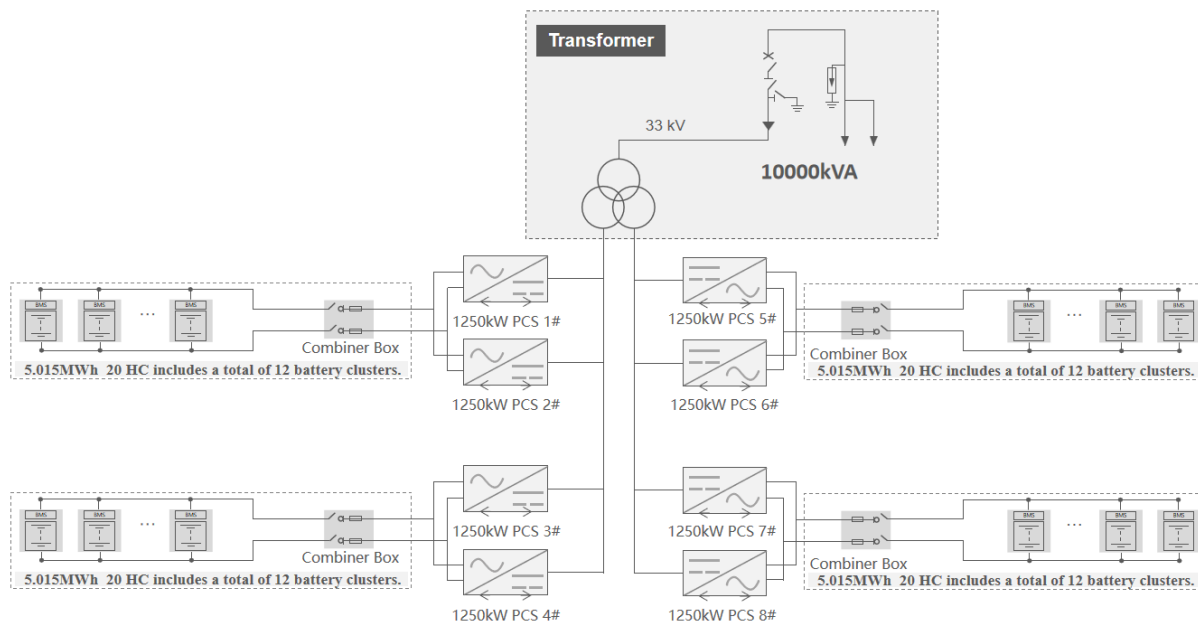


Figure Schematic diagram of a single 10MW / 20MWh @ 33kV energy storage unit

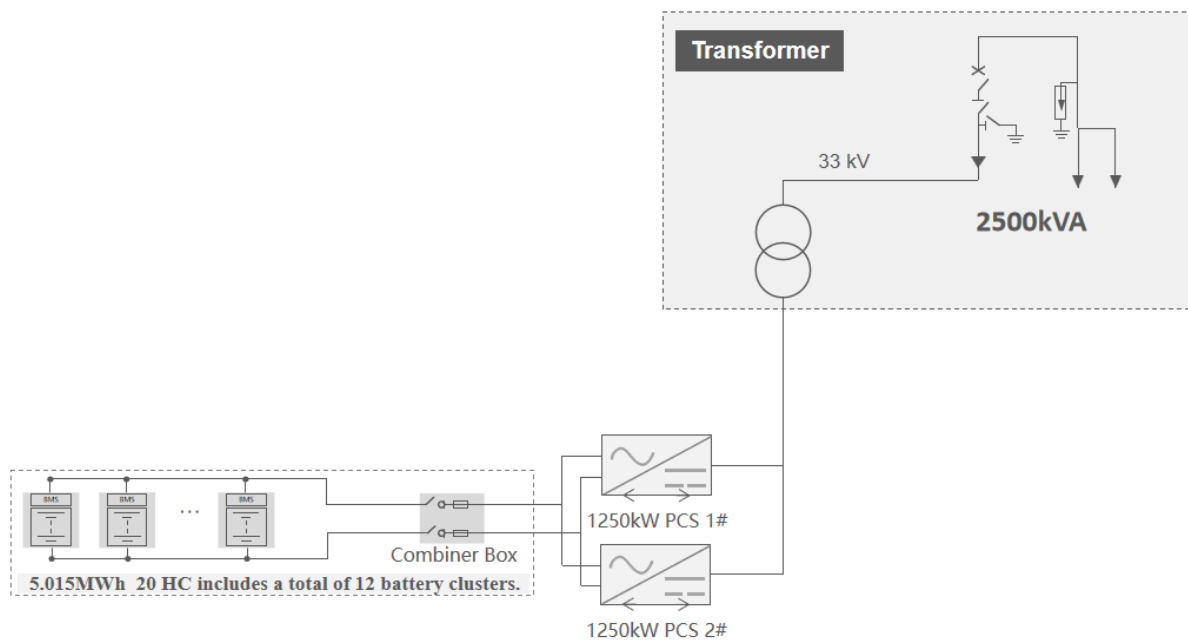


Figure Schematic diagram of a single 2.5MW / 5MWh @ 33kV energy storage unit

1.4 Configuration List (Supply Scope)

50 MW/ 100 MWh Liquid-Cooling Energy Storage System Configuration List (Initial)				Cyprus	
No.	Item	Specifications	Unit	Qty	Remark
1	1500V Liquid Cooling Battery Container	20 HC 5.015MWh lithium iron phosphate battery system	set	24	
1.1	Battery	5.015MWh (314Ah, 1P104S4S*12P, 0.5P, including BMU, BCU, BAU three-level battery management, PDU, battery rack and system accessories, etc.)	set	1	- EVE battery cell
1.2	BMS	Including BMU, BCU, BAU three-level architecture	set	1	
1.3	Control Busbar Cabinet	12 in 1, including UPS, Switches, Circuit Breaker, etc.	set	1	
1.4	Temperature Control System	Liquid cooling unit	set	1	
1.5	Fire Suppression System	Pack and system level aerosol fire suppression, system level combustible gas detectors + explosion-proof ventilation system, water fire suppression system	set	1	
1.6	Container	Dimensions: (W*H*D): 6058 × 2896 × 2438mm	pcs	1	- C5-M - IP55 rating - ≤2000m - Auxiliary power required (From Auxiliary transformer and UPS in PCS & MV skids) - Paint Colour: Door & Container body RAL9003; Bottom RAL9005
2	PCS&MV Skid Container	10MW	set	6	
2.1	PCS	BCS1250K-C-HUD	set	8	- Kehua - C4-M - IP65 rating

2.2	Step-up Transformer	10000kVA/ 33-0.69kV, oil immersed	set	1	-10000kVA@40℃ -Dy11y11 -Al/Al -Ecodesign Tier2 - Transformer body: IP68, Other parts: IP54 - Mineral oil
2.3	Auxiliary Transformer	150kVA/ 0.69-0.38kV, dry type	set	1	- Auxiliary power for PCS&MV Skid and Battery Container
2.4	Ring Main Unit	SF6	set	1	-Chinese Brand -40.5kV -SF6 -630A
2.5	Enclosure and Aux	Dimensions: (W*H*D): 12192 × 2896 × 2438mm Including internal cable accessories, auxiliary power distribution, etc.	pcs	1	- C4-M - IP54 rating - ≤2000m - Standard for no supply of oil retention tank

50 MW/ 100 MWh Liquid-Cooling Energy Storage System Configuration List (Augmentation from the 3 rd year)					Cyprus
No.	Item	Specifications	Unit	Qty	Remark
1	1500V Liquid Cooling Battery Container	20 HC 5.015MWh lithium iron phosphate battery system	set	5	
1.1	Battery	5.015MWh (314Ah, 1P104S4S*12P, 0.5P, including BMU, BCU, BAU three-level battery management, PDU, battery rack and system accessories, etc.)	set	1	- EVE battery cell
1.2	BMS	Including BMU, BCU, BAU three-level architecture	set	1	
1.3	Control Busbar Cabinet	12 in 1, including UPS, Switches, Circuit Breaker, etc.	set	1	
1.4	Temperature Control System	Liquid cooling unit	set	1	
1.5	Fire Suppression System	Pack and system level aerosol fire suppression, system level combustible gas detectors +	set	1	

		explosion-proof ventilation system, water fire suppression system			
1.6	Container	Dimensions: (W*H*D): 6058 × 2896 × 2438mm	pcs	1	- C5-M - IP55 rating - ≤2000m - Auxiliary power required (From Auxiliary transformer and UPS in PCS & MV skids) - Paint Colour: Door & Container body RAL9003; Bottom RAL9005
2	PCS&MV Skid Container	10MW	Set	1	
2.1	PCS	BCS1250K-C-HUD	Set	8	- Kehua - C4-M - IP65 rating
2.2	Step-up Transformer	10000kVA/ 33-0.69kV, oil immersed	set	1	-10000kVA@40℃ -Dy11y11 -Al/Al -Ecodesign Tier2 - Transformer body: IP68, Other parts: IP54 - Mineral oil
2.3	Auxiliary Transformer	150kVA/ 0.69-0.38kV, dry type	set	1	- Auxiliary power for PCS&MV Skid and Battery Container
2.4	Ring Main Unit	SF6	set	1	-Chinese Brand -40.5kV -SF6 -630A
2.5	Enclosure and Aux	Dimensions: (W*H*D): 12192 × 2896 × 2438mm Including internal cable accessories, auxiliary power distribution, etc.	pcs	1	- C4-M - IP54 rating - ≤2000m - Standard for no supply of oil retention tank
3	PCS&MV Skid Container	2.5MW	Set	1	

3.1	PCS	BCS1250K-C-HUD	Set	2	- Kehua - C4-M - IP65 rating
3.2	Step-up Transformer	2500kVA/ 33-0.69kV, oil immersed	set	1	-2500kVA@40°C -Dy11 -Al/Al -Ecodesign Tier2 - Transformer body: IP68, Other parts: IP54 - Mineral oil
3.3	Auxiliary Transformer	75kVA/ 0.69-0.38kV, dry type	set	1	- Auxiliary power for PCS&MV Skid and Battery Container
3.4	Ring Main Unit	SF6	set	1	-Chinese Brand -40.5kV -SF6 -630A
3.5	Enclosure and Aux	Dimensions: (W*H*D): 6058 × 2896 × 2438mm Including internal cable accessories, auxiliary power distribution, etc.	pcs	1	- C4-M - IP54 rating - ≤2000m - Standard for no supply of oil retention tank

Table Configuration List

Note:

1. NOT included: the power cables and the communication cables between the PCS&MV Skid, the EMS and the battery container.
2. The nameplate capacity is 60MW/120MWh initially, and a 12.5MW/25MWh for augmentation.
3. Product applicable altitude: <2000m; Ambient temperature range: -20°C~40°C.
4. Single battery container noise: 75dB at 1m, single PCS noise: 75dB at 1m.
5. Grid support: Active & reactive power control, grid following.
6. Preliminary proposal: for reference only. Final execution shall follow project realities; equipment selection is subject to project-based adjustment.

2 Lithium Iron Phosphate Battery Container

2.1 Functions and Features

This container energy storage system is mainly used for storing and transferring energy, which supports battery status monitoring and management, wide application such as grid peak regulation, frequency regulation, grid-connected access of new energy power stations, etc.

This Liquid cooling energy storage system is a highly integrated product, which integrates battery system, battery management system, thermal management system, fire suppression system, power supply and distribution system, monitoring system, environmental control system, and cable routing, etc., and has the characteristics of high safety and reliability, easy installation, high energy density and long cycle life.

2.2 Product Appearance



Figure Appearance of 5.015MWh container (Reference)

2.3 System Description

2.3.1 System Specification

Table Parameters of energy storage system

Item	Specification
Battery capacity (BOL) at DC side	5015kWh

Battery type		LFP
Configuration		12P416S
Rated energy/ kWh		5015
Battery voltage range/ V		1164.8 ~ 1497.6
Duration/ h		≥2
Communication		Ethernet / CAN / RS485
Thermal Management		Liquid cooling
Dimension (W * D * H)/ mm		6,058 * 2,438 * 2,896
Weight/ T		~ 41.5
Operation temperature/ °C	Charge	0 - 55
	Discharge	-30 - 55
Humidity (storage/operating)		<95% RH (non-condensing)
IP level		IP55
Certificates		IEC 62619/ IEC 63056/ IEC 61000/ IEC 62477-1/ UN 3536

2.3.2 Single Line Diagram

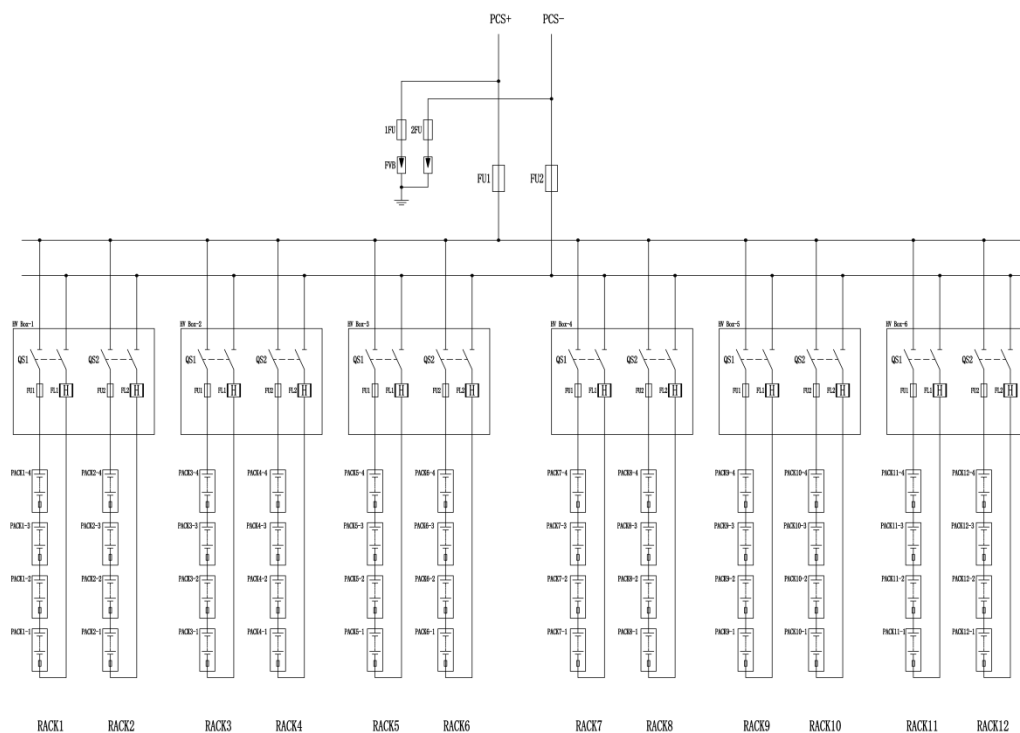


Figure Preliminary Single Line Diagram (Reference)

2.3.3 System Architecture

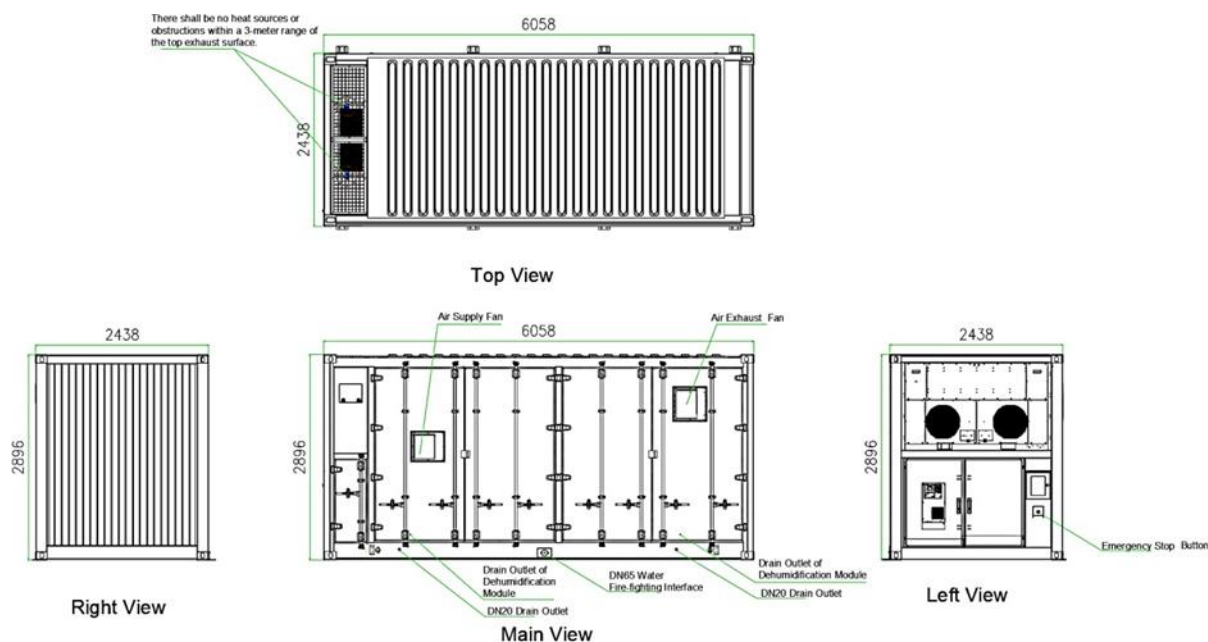


Figure The structure diagram of the container system (Reference)

2.3.4 System Components

Table Components of the container system

No.	Sub-parts	Components	Quantity	Remark
1	Battery container	20HC	1	Including distribution and lighting system
2	Battery system	Battery pack	48	1P104S(LFP 314Ah Cell), BMU
		Battery string switchgear unit	12	Including BCMU, breaker, fuse, contactor, current sensor, etc.
		Power Distribution Cabinet	1	Including BAMS, disconnecter, UPS
3	High Voltage Box	BCU, Circuit breaker, CT, Contactor	6	Control box of each two strings
4	BMS	BMS system	1	3-level architecture Including BMU, BCMU, BAMU
5	Liquid cooling system	Liquid cooling machine	1	Cooling capacity: 60kW (for 0.5C system)

		Liquid cooling pipes	1	Three-stage pipes
6	Fire suppression system	Aerosol	6	For container level
			96	For pack level
		Fire control panel	1	
		Smoke detector	2	Spot-type photoelectric
		Heat detector	2	Spot-type
		Gas detector	2	H ₂
		Air intake fan	1	
		Explosion-proof exhaust fan	1	
		Water firefighting system	1	

2.4 Component Description

2.4.1 Battery Cell

Table Specification of battery cell

Item		Specification
Battery type		LFP
Capacity/ Ah		314
Nominal voltage/ V		3.2
Voltage range/ V		2.5 - 3.65
Rated energy/ Wh		1004.8
Charge/Discharge Current	Standard	0.5P
Energy density/ Wh/kg		180
Life cycle (25°C, 90%DOD, 70%SOH)		6,000 (0.5P/0.5P)
Dimension (W * D * H)/ mm		(207±1.0) * (174 ±1.0) * (71.7±2.0)
Weight/ Kg		5.6±0.3
Operation temperature/ °C	Charge	0 - 50
	Discharge	-30 - 50
Certificates		UL 1973 / UL 9540A / IEC 62619 UN 38.3 / GB/T 36276 / ROHS / Bankability

2.4.2 Battery Pack

Table Parameters of battery pack

Item		Specification
Pack type		1P104S
Capacity/ Ah		314
Rated energy/ kWh		104.499
Configuration		1P104S
Nominal voltage/ V		332.8
Voltage range/ V		291.2 - 374.4
Duration/ h		≥2
Operation temperature/ °C	Charge	0 - 50
	Discharge	-30 - 50
Dimension (W * D * H)/ mm		(790±5) * (2180±5) * (252±5)
Weight/ Kg		~690±10
IP level		IP67
Thermal management		Liquid cooling
Certificates		IEC 62619/63056/EN IEC 61000-6-2/61000-6-4



Figure The diagram of battery pack (Reference)

2.4.3 Battery String

Table Parameters of battery string

Item	Specification
------	---------------

String type		1P416S
Battery type		LFP
Capacity/ Ah		314
Rated energy/ kWh		417.996
Configuration		1P416S
Nominal voltage/ V		1331.2
Voltage range/ V		1164.8 – 1497.6
Duration/ h		≥2
Operation temperature/ °C	Charge	0 - 50
	Discharge	-30 - 50
Thermal management		Liquid cooling
Certificates (All finished)		IEC 62619/63056/EN IEC 61000-6-2/61000-6-4

2.4.4 Power Distribution Cabinet

The power distribution cabinet is used for auxiliary power distribution, convergence and control of power distribution in the system, and plays a role in protecting the normal operation of the entire system.

The power distribution cabinet provides auxiliary power for the following items.

- Battery System
- Fire extinguishing system
- Air conditioner
- Liquid cooling unit
- Socket
- Other auxiliary power supply equipment

2.4.5 High Voltage Box

The overseas 1500V/200A dual-path integrated switch energy storage high-voltage box is an intermediate unit connecting the battery cluster and the energy storage converter. The energy

storage high-voltage box has functions such as battery cluster voltage collection, battery cluster current collection, battery cluster circuit contactor control and protection.

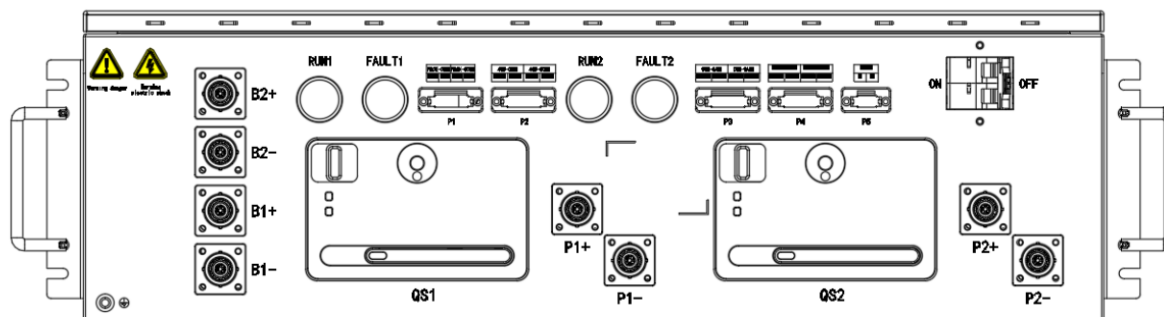


Figure High voltage box panel indication diagram (Reference)

Table Specification of High Voltage Box

NO.	Type	Effect
1	B+	Battery cluster input positive terminal
2	B-	Battery cluster input negative terminal
3	P+	PCS input positive terminal
4	P-	PCS input negative terminal
5	QS1	Battery Cluster Switch
6	QS2	Battery Cluster Switch
7	P1	Power the main control module
8	P2	Dry Contact
9	P3	Communication/debugging CAN
10	P4	Daisy chain/485 communication

2.4.6 BMS

Table Technical parameters and indicators required for the battery management system

Level	Configuration	Functionality
Level 1	BMU (Battery Module Unit) *1 per module	1. Detects individual cell voltage and temperature, compiles the collected data, and reports it. 2. Acts as the balancing unit for cell voltage, performing autonomous balancing or following commands from higher-level units. 3. Conducts real-time self-diagnosis of sampling and balancing circuits, reporting results in real-time. 4. Communicates with the higher-level BCMS through Daisy Chain communication.
Level 2	BCMU (Battery string Management Unit) *1 per battery string	1. Collects temperature, voltage, current, and SOC information uploaded by battery modules within the string. 2. Calculates battery SOH based on calibration operations. 3. Manages the normal operation and alarm information of the battery string, controlling the opening and closing of the string's DC switch as needed. 4. Issues balancing commands based on the voltage status within the string. Issues insulation detection commands as per higher-level instructions. 5. Uploads battery string parameters and switch status to the BAMS through Ethernet or CAN communication, and receives control instructions from BAMS.
Level 3	BAMS (Battery Stack Management Unit) *1	1. Collects and manages BCMS information from the second level through the switch, overseeing voltage, current, temperature, and SOC of lower levels. 2. Manages the direction of the energy flow of the storage system in coordination with the EMS, SCADA, and PCS management systems. 3. Monitors bus voltage, total current, and bus insulation. 4. Consolidates overall alarm and protection statuses and decides whether to control the main circuit breaker for protection. 5. Calculates the SOC of connected batteries and calibrates battery capacity based on the cycle status. 6. Transmits data to the touchscreen and responds to local control commands issued from the touchscreen.

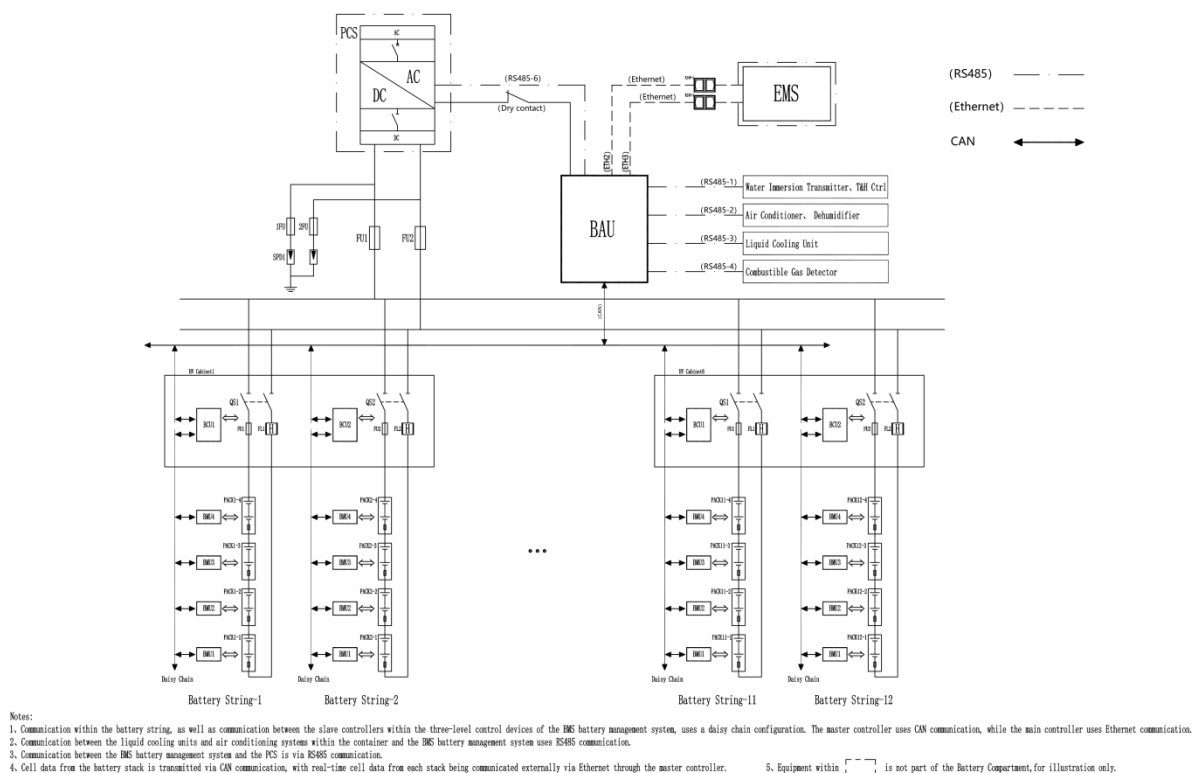


Figure Communication framework of BMS (Reference)

Table BMS electrical parameters

Item		Parameter	Remarks
*Balancing Method		Passive	
*Balancing Current		$\geq 100\text{mA}$	
*Balancing Efficiency		/	
*Power Consumption	Cell Monitoring Module Power Consumption	$\leq 2\text{W}$	
	BMS Control Module Power Consumption	$\leq 3\text{W}$	
	BMS Display and Control Module Power Consumption	$\leq 15\text{W}$	

2.4.7 Liquid Cooling System

The battery part of the system adopts liquid cooling technology to dissipate heat. Liquid cooling technology brings less loss and better temperature uniformity. The liquid cooling system mainly comprises of pipes, pumps, heat exchangers and compressors. The coolant of the system is

mixed solution of ethylene glycol and water. The coolant flows from the water inlet pipe to the 6 main longitudinal branches. Each main branch is divided into 8 branches and flows to the liquid cooling pack to cool / heat the cell. After flowing out of the pack, it is summarized to the return pipe through the branch pipe and returned to the liquid cooling unit.

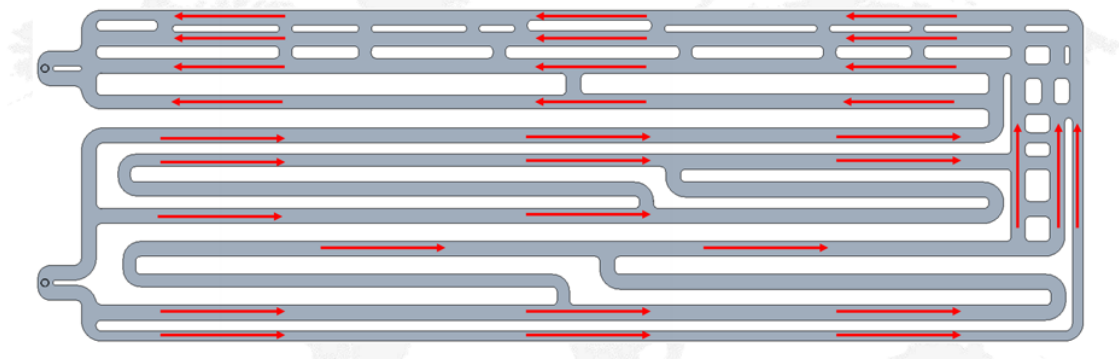


Figure Flow direction of coolant in pack (Reference)

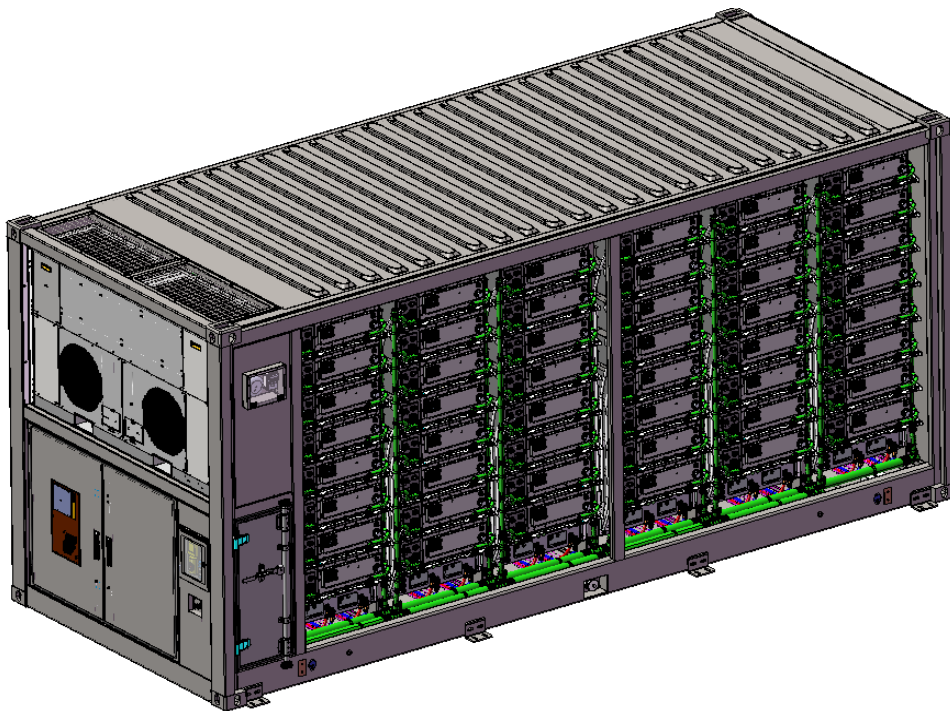


Figure Liquid cooling design of system (Reference)

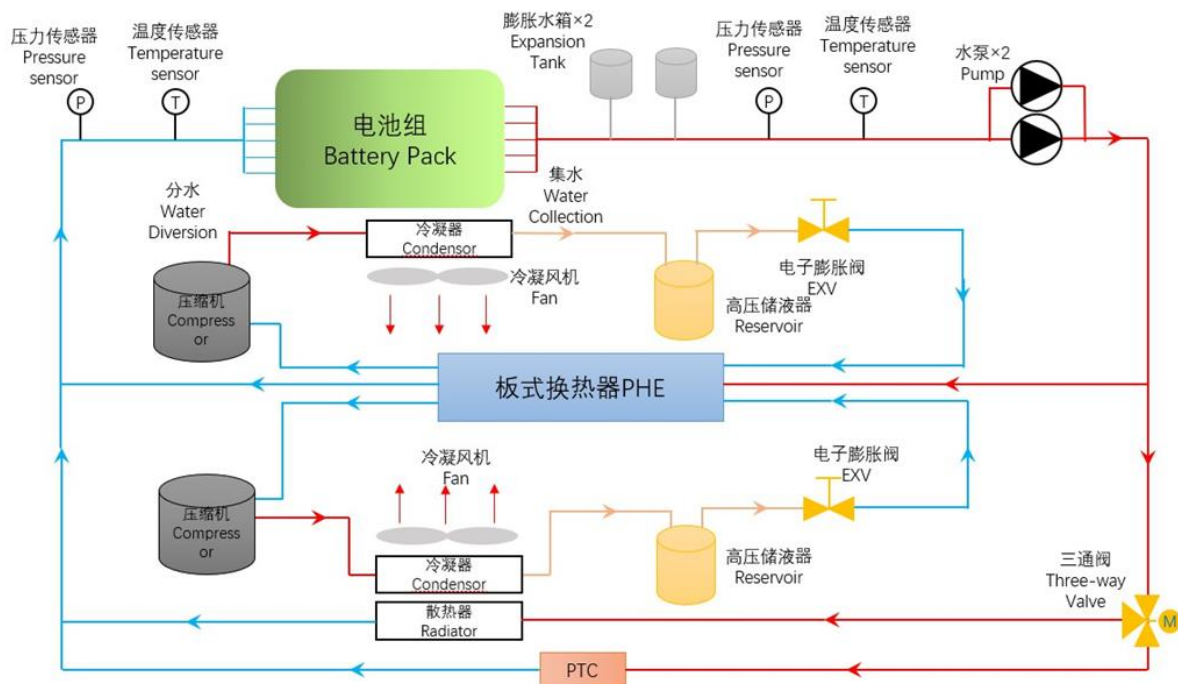


Figure System schematic (Reference)

Table The liquid cooling unit main technical parameters

Model	BTMS-600-ES
Cooling Capacity / kW	60(The ambient temperature is 45 °C and the outlet water temperature is 20 °C)
Heating Capacity / kW	16
Total Power Supply / V	480V (323~552V)
Frequency / Hz	50/60
Phase	3
No. of Wires	5
Total A – Cooling / A	43
Maximum Operating Current / A	60
Total A – Heating / A	26
Refrigerant, Type	R513a
Amount / oz	113*2
Design Pressure	-
High Side / psig (MPa)	319 (2.2)
Low Side / psig (MPa)	217 (1.5)
Operating Ambient Temperature / °C	-30~55
Coolant Outlet Temperature / °C	5~35(Cooling); <50(Heating)
Rate Of Flow / L/Min	500 (@150kPa battery side pressure drop)
Altitude / m	0~4000
IP	IP26(Entire machine) IP67(Electrical components)
Sound / (A)	≤75(JB/T 4330)

Cabinet Outer Enclosure	
Length / mm (max)	2100
Depth / mm	550
Height / mm (max)	1230
Thickness / mm (min)	2.5
Corrosion Protection (Method)	Coating / Layer C5M
Material	Aluminum
Weight / kg	≤410
Color	RAL9003

2.4.8 Fire Suppression System

The designs the container fire protection solution with automatic fire protection system + explosion-proof system + water firefighting system (see topology and logic diagram of the fire protection system of the container of electrochemical energy storage), which strictly follows the relevant national standards and norms in order to ensure the scientificity, reasonableness and validity of the scheme, comprehensively safeguard fire protection of the electrochemical energy storage system, and ensure that the system can be safely run and highly effective response to fire accidents.

(1) Ventilation systems:

When combustible gas is generated from thermal runaway of the battery cell in the container, the combustible gas detector will immediately activate the ventilation system to ventilate the area when it detects that the concentration of combustible gas has reached a preset value, so as to prevent the risk of explosion when the combustible gas reaches the explosive concentration and encounters external energy.

(2) Automatic fire suppression systems:

When a fire occurs in the container, the smoke detector and the heat detector detect the fire and transmits the signal to the gas fire suppression controller, which activates the aerosol in the container to start the fire extinguishing.

(3) Water firefighting system:

Due to environmental or other factors, the automatic fire suppression system fails or the fire

spreads to become a fire, professionals need to connect the water from the municipal fire hydrant to the water fire interface on the container to start the emergency water supply system.

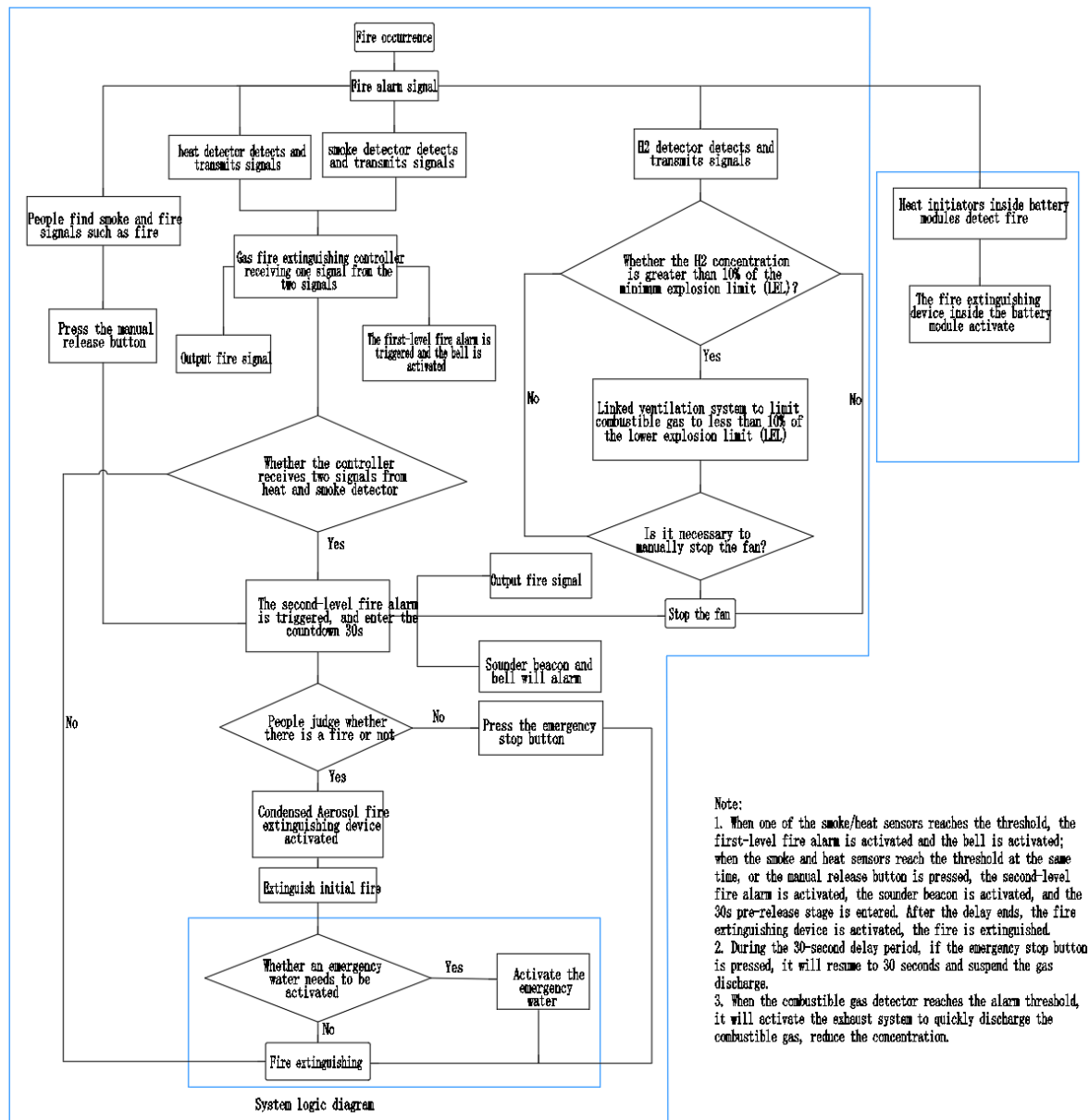


Figure Logic diagram of the container BESS and fire protection system (Reference)

3 PCS & MV skid Container

Energy storage systems are a vital component in today's grid power, providing a multitude of crucial functions to the grid. In alignment with the functional requirements of this project, our energy storage system incorporates converter equipment, ensuring both high efficiency and reliability of the system.

3.1 General Technical Specification

3.1.1 Primary Diagram of System

The complete set of 40ft. 10 MW PCS & MV skid container adopted an integrated form of 1250 kW energy storage PCS $\times$ 8, one 10000 kVA oil-immersed step-up transformer and one communication power cabinet into one equipment tray and reaches IP54 protection level. 2.5MW PCS & MV skid container adopted an integrated form of 1250 kW energy storage PCS $\times$ 2, one 2500 kVA oil-immersed step-up transformer and one communication power cabinet into one equipment tray and reaches IP54 protection level.

The topology diagram is as follows:

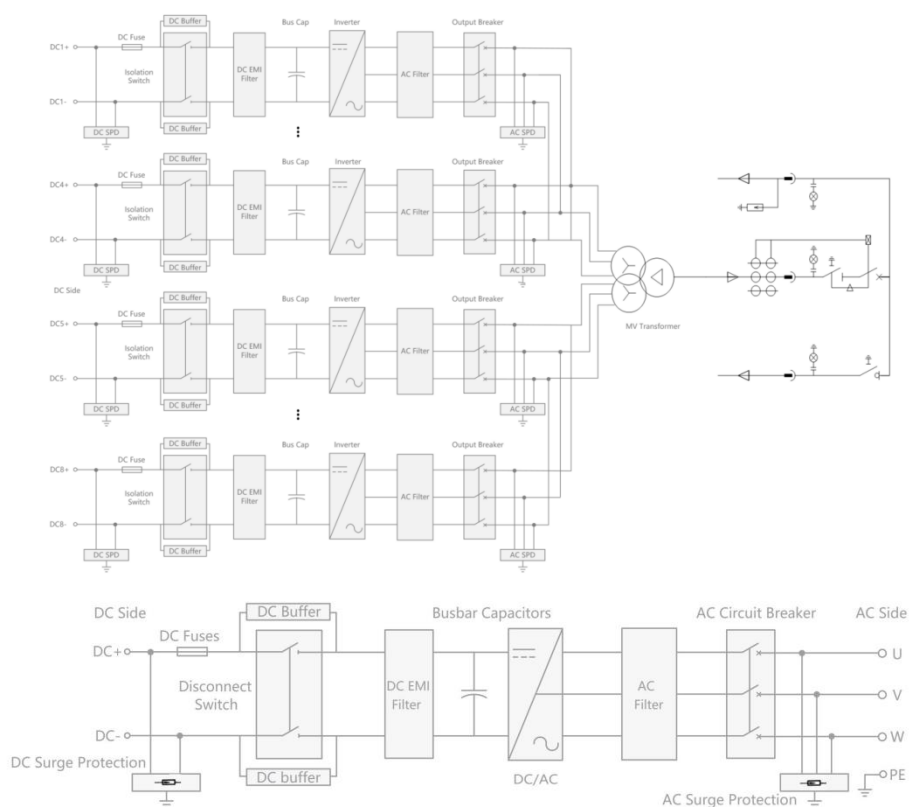


Figure PCS topology diagram (reference)

3.1.2 Complete Equipment Layout and Construction Specifications

The equipment layout is as follows:



Figure The appearance (for reference only, adjustable according to specific projects)

During foundation construction, when multiple container type medium voltage PCS are placed in parallel, the container spacing should not be less than 3m, and 4m or more is recommended to ensure the heat dissipation of the containers.

Table Parameters of PCS & MV skid

Type	BCS10000K-C-HUD/T8
DC Input	
Max. DC voltage	1500Vdc
DC voltage range	1060~1500Vdc
Max. DC current	1402A*8
Soft start	Yes
LV AC Output, on grid	

Rated AC output power	10000kW
Max. AC output power	11000kVA
Rated grid-tied voltage	690Vac
Grid voltage range	-15%~10%(settable)
Rated grid frequency	50/60Hz
Max. output current	9204A
Power factor/Adjustable range	>0.99(rated power)/-1(leading)~1(lagging)
Range of reactive power	-100 % ~ 100 %
THDi	<3% (rated power)
LV AC Output, off grid	
Rated AC output voltage	690Vac
Output voltage accuracy	1%
Max. output current	9204A
THDu	<3% (linear load)
Rated output frequency	50/60Hz
PCS Efficiency	
Max. efficiency	≥99%
Transformer	
Rated power	10000kVA
Voltage transformation ratio	0.69/33kV
Isolation mode	Oil-immersed transformer
Transformer vector	Dy11-y11
General Data	
IP rating	IP65(PCS) IP54(Skid)
Operation temperature	-35°C~60°C (> 45°C derating)
Relatively humidity	0~100%(no-condensing)
Cooling type	Intelligent air cooling
*Dimensions(W*H*D)	12192*2896*2438mm (40ft)
*Weight	≤40000kg
Altitude	4000m (>2000m derating)
Display	LED
Communication Protocol	Modbus-RTU / Modbus-TCP / IEC61850 /IEC104
Communication Interface	RS485/CAN/Ethernet
Compliance	EN/IEC 62477-1, EN/IEC 61000-6-2, EN50549-2/10, IEC62116, IEC61727, IEC60068-2-1/2/14/30
Note:	
*Reference only, subject to project product drawings	
*The specifications are subject to change without prior notice.	

Type	BCS2500K-C-HUD/T2
DC Input	

Max. DC voltage	1500Vdc
DC voltage range	1060~1500 Vdc
Max. DC current	1403A*2
Soft start	Yes
LV AC Output, on grid	
Rated AC output power	2500kW
Max. AC output power	2750kVA
Rated grid-tied voltage	690Vac
Grid voltage range	-15%~10%(settable)
Rated grid frequency	50/60Hz
Max. output current	2301A
Power factor/Adjustable range	>0.99(rated power)/-1(leading)~1(lagging)
Range of reactive power	-100 % ~ 100 %
THDi	<3% (rated power)
LV AC Output, off grid	
Rated AC output voltage	690Vac
Output voltage accuracy	1%
Max. output current	2301A
THDu	<3% (linear load)
Rated output frequency	50/60Hz
PCS Efficiency	
Max. efficiency	≥99%
Transformer	
Rated power	2500kVA
Voltage transformation ratio	0.69/35kV
Isolation mode	Oil-immersed transformer
Transformer vector	Dy11
General Data	
IP rating	IP65(PCS) IP54(Skid)
Operation temperature	-35°C~60°C (> 45°C derating)
Relatively humidity	0~100%(no-condensing)
Cooling type	Intelligent air cooling
*Dimensions(W*H*D)	6058*2896*2438 (20ft)
*Weight	<27000kg
Altitude	4000m (>2000m derating)
Display	LED
Communication Protocol	Modbus-RTU / Modbus-TCP / IEC61850 / IEC104
Communication Interface	RS485/CAN/Ethernet
Compliance	EN/IEC 62477-1, EN/IEC 61000-6-2, EN50549-2/10, IEC62116, IEC61727, IEC60068-2-1/2/14/30
Note:	
*Reference only, subject to project product drawings	

*The specifications are subject to change without prior notice.

3.2 PCS Product Specification

3.2.1 Appearance of PCS



Figure Appearance of PCS (Reference)

3.2.2 PCS Technical Parameters

Table Parameters of PCS

Type designation	BCS1250K-C-HUD
DC Input	
Max. DC voltage	1500Vdc
DC voltage range	1060~1500Vdc
Max. DC current	1403A
Soft start	Yes
LV AC Output, on grid	
Rated AC output power	1250kW
Max. AC output power	1375kVA
Rated grid-tied voltage	690Vac
Grid voltage range	-15%~10%(settable)
Rated grid frequency	50Hz
Max. output current	1150.6A
Power factor/Adjustable range	>0.99(rated power)/-1(leading)~1(lagging)
Range of reactive power	-100 % ~ 100 %

THDi	<3% (rated power)
LV AC Output, off grid	
Rated AC output voltage	690Vac
Output voltage accuracy	1%
Max. output current	1150.6A
THDu	≤3% (linear load)
Rated output frequency	50Hz
PCS Efficiency	
Max. efficiency	≥99%
General Data	
Isolation mode	None
IP rating	IP65(PCS)
Operation temperature	-35°C~60°C (> 50°C derating)
Relatively humidity	0~100%(no-condensing)
Cooling type	Intelligent air cooling
*Dimensions(W*H*D)	735*2135*1300 mm
*Weight	About 950kg
Altitude	4000m (>2000m derating)
Display	LED
Communication Protocol	Modbus-RTU / Modbus-TCP / IEC61850 / IEC104
Communication Interface	RS485/CAN/Ethernet
Compliance	EN/IEC 62477-1, EN/IEC 61000-6-2, EN50549-2/10, IEC62116, IEC61727, IEC60068-2-1/2/14/30
Note: *Reference only, subject to project product drawings *The specifications are subject to change without prior notice.	